August 2026

Mahdi Mokhtarzadeh

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RESEARCH INTERESTS:

- Operations Research and Management
- Quantum Computing in Optimization and Machine Learning
- Probabilistic Reasoning and Causal Inference
- Production Scheduling, *Seru* Production Systems, and Assembly Line Balancing
- Intelligent Quality Management, Failure Analysis, and Reliability Engineering
- Industry 4.0 and 5.0, Smart Manufacturing, and Human-Robot Collaboration

PROFESSIONAL SKILLS:

- Statistics and Data Analysis
    - Bayesian *Statistics and Causality*
    - Supervised and Unsupervised Machine Learning
    - Causal Machine Learning
- Mathematical Programming and Optimization
    - Constraint Programming
    - Benders and Dantzig Wolfe Decompositions
    - Nonlinear Optimization
    - Metaheuristic Algorithms

EDUCATION:

Apr. 2022 – Dec. 2026	Ph.D., Industrial Engineering, Ghent University, Ghent, Belgium, Thesis Title: *Probabilistic Reasoning for Quality Management in Industry 4.0*, Supervisor: Prof. Sidharta Gautama
Sep. 2017 - Jul. 2020	M.Sc., Industrial Engineering, University of Tehran, Tehran, Iran, GPA: 19.21/20, Thesis Title: *Production Planning of Mixed Model Assembly Lines by Considering Human Factors and Customer Satisfaction*, Supervisor: Prof. Masoud Rabbani
Sep. 2011 - Jan. 2015	B.Sc., Industrial Engineering, University of Hormozgan, Bandar Abbas, Iran, GPA: 16.24/20

PROFESSIONAL EXPERIENCE:

Apr. 2022 – Present	Research Assistant, Department of Industrial Systems Engineering and Product Design, Ghent University, Ghent, Belgium Project Title: *Assembly Quality Management Using System Intelligence*, Supervisor: Prof. Sidharta Gautama
Sep. 2026 – Dec. 2026	Research Visit, GEMTEX Textile Research Laboratory, ENSAIT, Lille University, Lille, France Project Title: Towards Human-Centered Decision Support Engineering for Intelligent Textile Manufacturing, Supervisor: Prof. Kim-Phuc Tran
Oct. 2018 – Jul. 2021	Teaching Assistant, University of Tehran, Tehran, Iran, Main Lecturer: Prof. Masoud Rabbani *Advanced Production Planning* (Oct. 2018 - Jul. 2019) *Advanced Engineering Economics* (Feb. 2021 - Jul. 2021)

LANGUAGE SKILLS:

Persian: *Native*

English: *Professional Proficiency*

TOEFL: 102, Reading: 26, Listening: 27, Speaking: 22, Writing: 27 (Nov. 2024)

Dutch: *Intermediate* (B1)

SELECTED PUBLICATIONS:

Peer-Reviewed:

1. **Mokhtarzadeh M.**, Rodríguez-Echeverría J., Gautama S. (2026). pFMEA-based Decision-Support Systems for Quality Management: Case Study. *29th International Conference on Information Fusion*.
2. **Mokhtarzadeh M.**, Rodríguez-Echeverría J., Gautama S. (2025). A Framework for Integrated Data-Supported Failure and Root Cause Analysis. *Reliability Engineering & System Safety*, 269, 112099.
3. **Mokhtarzadeh M.**, Rodríguez-Echeverría J., Semanjski I., Gautama S. (2025). Hybrid Intelligence Failure Analysis for Industry 4.0: A Literature Review and Future Prospective. *Journal of Intelligent Manufacturing*, 36, 2309–2334.
4. **Mokhtarzadeh M.**, Rodríguez-Echeverría J., Zeren Z., Van Noten J., Gautama S. (2024). Pure Data-Driven Machine Learning Challenges for pFMEA: A Case Study, *IFAC-PapersOnLine*, 58 (19), 658-663.
5. Farsi A., **Mokhtarzadeh M.**, Rabbani M., Manavizadeh N., Ghasempour Anaraki M. (2024). Using Parallel Metaheuristics to Solve a Parallel U-shaped Robotic Mixed-model Assembly Line Balancing and Sequencing Problem. *Soft Computing*, 28, 12603–12621.
6. Stecke K., **Mokhtarzadeh M.** (2022). Balancing Collaborative Human-Robot Assembly Lines to Optimize Cycle Time and Ergonomic Risk. *International Journal of Production Research,* 60 (1), 25-47.
7. **Mokhtarzadeh M.**, Tavakkoli-Moghaddam R., Triki C., Rahimi Y. (2021). A Hybrid of Clustering and Meta-Heuristic Algorithms to Solve a *p*-Mobile Hub Location–Allocation Problem with the Depreciation Cost of Hub Facilities. *Engineering Applications of Artificial Intelligence*, 98, 104121.
8. Triki C., Mahdavi Amiri M., Tavakkoli-Moghaddam R., **Mokhtarzadeh M.**, Ghezavati V. (2021). A Combinatorial Auction-Based Approach for the Ridesharing in a Student Transportation System. *Networks*, 78, 229–247.
9. **Mokhtarzadeh M.**, Rabbani M., Manavizadeh N. (2021). A Novel Two-Stage Framework for Reducing Ergonomic Risks of a Mixed-Model Parallel U-Shaped Assembly-Line. *Applied Mathematical Modelling*, 93, 597-617.
10. **Mokhtarzadeh M.**, Tavakkoli-Moghaddam R., Vahedi-Nouri B., Farsi A. (2020). Scheduling of Human-Robot Collaboration in Assembly of Printed Circuit Boards: A Constraint Programming Approach. *International Journal of Computer Integrated Manufacturing*, 33(5), 460-473.

Works in Progress:

1. Schmitt T., **Mokhtarzadeh M.**, Stecke K., Kovacs K. (2026). Digital and Quantum Steps towards Profit in Job Shop Scheduling.
2. **Mokhtarzadeh M.**, Rodríguez-Echeverría J., Gautama S. (2026). A Context-Aware Human-in-the-Loop Decision Support System for Failure Analysis in Smart Manufacturing.
3. **Mokhtarzadeh M.**, Stecke K. (2026). Competitive Joint Network Design and Pricing Problem for On-demand Drone Delivery.

CONFERENCES and SEMINARS:

2026, 29th International Conference on Information Fusion (FUSION 2026)
2025, Flanders Make Symposium 2025 - Where Science Meets Solutions
2025, 34th European Conference on Operational Research (EURO 2025)
2024, 18th IFAC Symposium on Information Control Problems in Manufacturing (INCOM 2024)

HONORS:

2022, Awarded Research Assistant Scholarship, Ghent University
2020, Admitted to Ph.D. Program in Operations Management, University of Texas at Dallas
2018-2019, Ranked First in the Class with a GPA of 19.21/20, University of Tehran
2017, Top 0.5% (54/10,000) in Iran National University Entrance Exam for a M.Sc. in Industrial Engineering

SELECTED COURSES:

Bayesian Statistics, Causal Machine Learning, Queuing Systems, Game Theory, Operations Research, and Discrete Optimization

COMPUTER SKILLS:

Programming Languages: MATLAB, Python, C, and C++
Optimization Software: IBM ILOG CPLEX, and Gurobi, D-Wave Quantum Annealing, and IBM Qiskit

HOBBIES and INTERESTS:

Sports: Chess, Swimming, and Table Tennis
Interests: Traveling, Hiking, History and Philosophy, and Literature (Poetry and Fiction)

REFERENCES:

- Prof. Sidharta Gautama, *Department of Industrial Systems Engineering and Product Design, Ghent University*, Ghent, Belgium, Sidharta.Gautama@UGent.be
- Prof. Thomas Schmitt, *Foster School of Business, University of Washington*, Seattle, WA, USA, glennsch@u.washington.edu